
More Agents, Better Results? Meta-Analysis of Multi-Agent v.s. Single-Agent LLM Systems

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Abstract

This analysis synthesizes comparisons (2023–2026) asks whether multi-agent LLM orchestration outperforms single-agent prompting on benchmark accuracy, and whether the advantage survives compute-matched comparison. The answer to the first question is: marginally, on average. The pooled effect is positive (log-OR = 0.45, 95% CI [0.23, 0.67]; primary accuracy-only $k = 37$: 0.46 95% CI [0.22, 0.71]) yet practically small (aggregate RD ≈ 3 pp, Cohen’s $h = 0.07$) and the prediction interval crosses zero (95% CI [−0.74, 1.64]). The answer to the second is: barely. Under compute parity ($k = 9$) the relative advantage declines $\sim 31\%$ (log-OR 0.31, 95% CI [0.006, 0.61]), with the lower bound essentially at zero. These findings are hypothesis-generating only: heterogeneity is extreme ($I^2 = 96.1\%$), Egger’s test detects funnel asymmetry ($p = .024$). Full analysis can be found in <https://github.com/VoteVeto2/Project-Meta-Analysis>.

1 Introduction

The multi-agent paradigm for LLM (Large Language Models) rests on an intuition: Two heads are better than one. Multiple LLM agents collaborating should exceed what a single agent achieves alone. Early work by Du et al. (2023) demonstrated that multi-agent debate improved factuality and reasoning in ChatGPT, while Liang et al. (2024) showed that multi-agent discussion encouraged divergent thinking. Frameworks such as AutoGen (Wu et al., 2023), MetaGPT (Hong et al., 2023), ChatDev (Qian et al., 2024) and Dynamic Flows in Claude Code (Anthropic, 2026) now orchestrate dozens of specialized agents in hierarchies and cooperative workflows.

Yet a skeptical counter-intuitive has emerged. Zhang et al. (2025) directly show that multi-agent debate often fails to outperform simple single-agent baselines such as chain-of-thought and self-consistency, even while consuming substantially more test-time compute. As frontier models like Claude-Opus-series and GPT-series improve (Anthropic, 2026) (OpenAI, 2026), the coordination overhead may outweigh diminishing marginal gains.

To our knowledge, no quantitative meta-analysis has yet synthesized these conflicting findings, though multi-agent LLM surveys and benchmarks exist (Guo et al., 2024). We address two research questions:

RQ1 (Primary). Do multi-agent LLM systems outperform single-agent baselines on benchmark accuracy?

RQ2 (Compute parity). Under matched compute budgets, does the multi-agent advantage survive?

2 Methods

2.1 Search strategy and limitations

We deployed six parallel search agents covering distinct topic clusters (Appendix C). Each agent queried arXiv, ACL Anthology, NeurIPS/ICML/ICLR proceedings, and Google Scholar, targeting papers from 2023–2026, yielding 153 records. **This is a convenience sample, not a registered systematic review.** No protocol was pre-registered; the six-agent approach introduces selection biases. Results should be treated as hypothesis-generating.

2.2 Screening

Population: Empirical 2023–2026 LLM studies.

Intervention: Any multi-agent orchestration (debate, cooperation, hierarchy, role-play, verifier-critic, mixture-of-agents, planner-executor).

Comparator: Single-agent baseline using the same backbone model.

Outcome: Accuracy or success-rate proportion on a named benchmark.

Study design: Within-study head-to-head comparison.

Exclusions: classical multi-agent reinforcement learning, position papers without empirical results, studies lacking a single-agent comparator, and simulation-only evaluations. After de-duplication (26 removed) and screening (21 excluded), 106 records remained; 41 reported extractable accuracy data (See Figure 1). Screening was performed by the author with AI assistance; no formal inter-rater reliability was computed.

2.3 Data extraction

For each included study, an AI extraction agent produced a structured row with study identifiers, backbone model, architecture type, task category, benchmark name, item counts for both conditions, compute-parity flag, and metric type. Full column definitions are in Appendix A.

2.4 Effect size

We computed log odds ratios (log-OR) as the primary effect size. Because item-level paired correctness was unavailable, MA and SA proportions were treated as independent binomial counts. Where arms are item-paired, ignoring the positive between-arm correlation overstates each study’s within-study variance and is therefore conservative. A 0.5 continuity correction was applied conditionally to zero-cell studies (Sweeting et al., 2004); none were present.

To aid practical interpretation, we report risk differences ($RD = p_{MA} - p_{SA}$). The RD is weighting-dependent: the aggregate (n -weighted) RD across all 41 studies is ≈ 3 pp, but the unweighted study mean is ≈ 8 pp (median 6 pp). The 41 comparisons comprise item-level accuracy (37), one win rate, and three resolve rates. Treating the four non-accuracy metrics as binomial proportions is a simplification, so the **primary specification is the 37 item-level-accuracy studies** ($k = 37$, log-OR = 0.46 [0.22, 0.71]); the full $k = 41$ is reported as a robustness inclusion.

2.5 Statistical model

The primary analysis used REML random-effects estimation (Viechtbauer, 2005) with the HKSJ adjustment (Hartung and Knapp, 2001; IntHout et al., 2014), applying $\max(1, s_{HK}^2)$ as a variance inflation floor (Röver et al., 2015). CIs use $t(k - 1)$ degrees of freedom; prediction intervals use $t(k - 2)$ (IntHout et al., 2016). DerSimonian-Laird served as a sensitivity check. Cochran’s Q uses fixed-effect weights (Borenstein et al., 2009, Ch. 16).

2.6 Risk of bias and publication bias

Of 41 studies, 25 (61%) were rated high risk of bias, predominantly due to uncontrolled compute. Only 10 of 41 rows are independently verified; among the 12 audited, 6 required correction (50%

error rate). Publication-bias diagnostics included funnel-plot inspection, Egger’s regression (Egger et al., 1997), Begg’s rank correlation (Begg and Mazumdar, 1994), and trim-and-fill (Duval and Tweedie, 2000); all are unreliable under extreme heterogeneity ($I^2 > 90\%$).

3 Results

3.1 Study flow

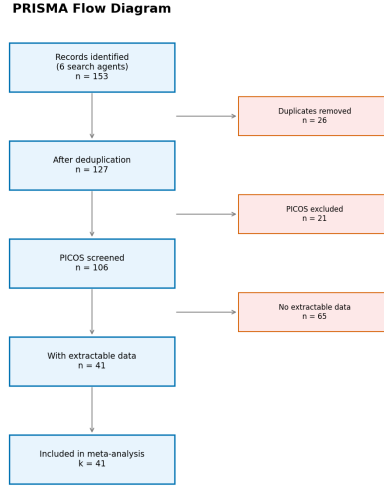


Figure 1. Flow diagram. Six search agents identified 153 records; after de-duplication (26), screening (21), and data-availability filtering (65), 41 studies were included. See Appendix D for the exclusion table.

3.2 Descriptive characteristics

The 41 comparisons span 2023–2026. Architectures: debate ($k = 20$), cooperation (9), hierarchy (6), verifier-critic (2), role-play (2), other (2). Task categories: code (11), other/mixed (11), medical (7), math (6), evaluation (3), general knowledge (3). Only 9 studies enforced compute parity. The complete extraction table is in Appendix A.

3.3 Pooled effect (RQ1)

Table 2. Primary pooled results ($k = 41$).

Estimator	log-OR	95% CI	95% PI	τ^2	I^2	Q (df = 40)
REML (HKSJ)	0.45	[0.23, 0.67]	[−0.74, 1.64]	0.34	96.1%	1016.6***
DerSimonian-Laird	0.41	[0.22, 0.61]	[−0.38, 1.21]	0.15	96.1%	1016.6***

*** $p < .001$. PI = prediction interval.

The pooled log-OR is 0.45 (95% CI [0.23, 0.67], $p < .001$) for the full sample and 0.46 [0.22, 0.71] for accuracy-only ($k = 37$), corresponding to OR ≈ 1.57 . The absolute gain is small: aggregate RD ≈ 3 pp (Cohen’s $h = 0.07$), versus an unweighted study mean of ≈ 8 pp ($h = 0.20$; Cohen, 1988). The aggregate figure is pulled down by large near-null general-knowledge benchmarks (e.g., MMLU, $n = 14,042$).

The main limitation is the quality and auditability of the extraction table. The pooled effect is positive but fragile: (1) The prediction interval [−0.74, 1.64] crosses zero. (2) Egger’s test detects funnel asymmetry ($p = .024$). (3) The absolute gain is small (aggregate Cohen’s $h = 0.07$). (4) 61%

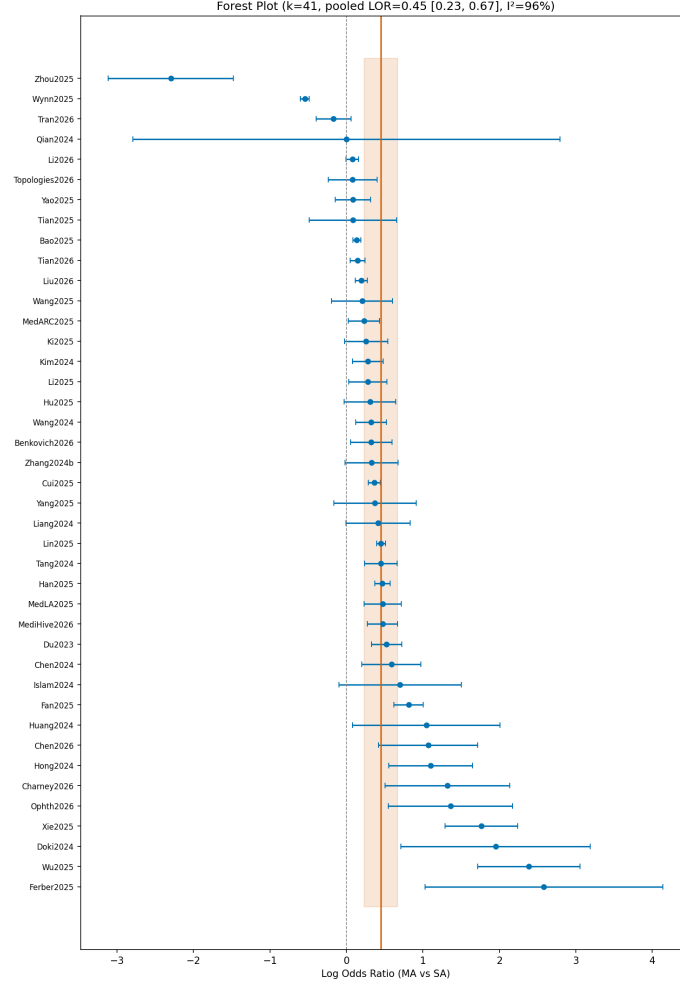


Figure 2. Forest plot. REML random-effects pooled log-OR = 0.45, 95% CI [0.23, 0.67].

of studies are high-RoB; 76% of rows lack independent verification. (5) $I^2 = 96.1\%$; the between-study SD ($\tau = 0.58$) exceeds the pooled estimate, so the pooled mean should not be used as a point prediction.

3.4 Compute parity (RQ2)

Under compute parity ($k = 9$), the relative effect declines $\sim 31\%$ (log-OR = 0.31, 95% CI [0.006, 0.61]); the CI lower bound barely escapes zero. On the absolute scale, however, the parity subset does **not** shrink: its aggregate RD is ≈ 5 pp (Cohen’s $h \approx 0.11$), larger than the full-sample ≈ 3 pp, because parity studies exclude the large near-null general-knowledge benchmarks. The relative multi-agent advantage is fragile under fair compute comparison, while the absolute gain stays small but is subset-dependent.

3.5 Verified subset as credibility anchor

The source-located/verified subset ($k = 10$) yields log-OR = 0.35 [0.19, 0.51], notably lower than the full sample, with sharply reduced heterogeneity ($\tau^2 = 0.01$ v.s. 0.34). The lower estimate warns that unverified rows may contain extraction errors that inflate both the pooled effect and the apparent between-study variation.

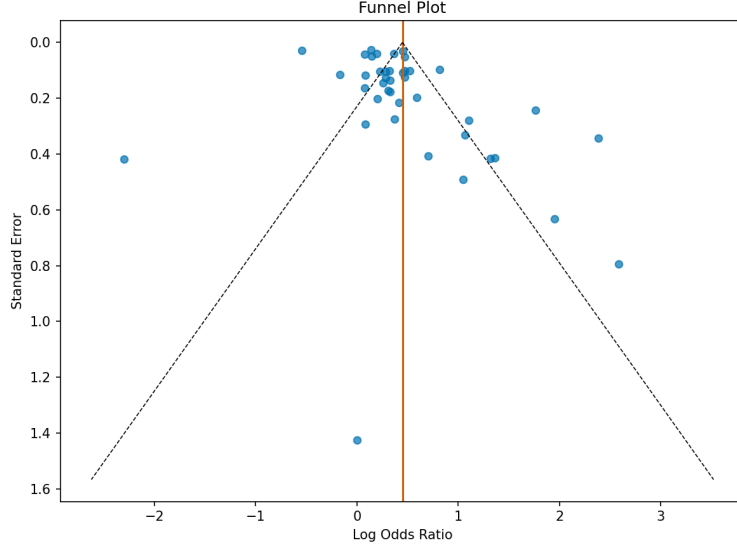


Figure 3. Funnel plot. Egger’s test detects significant asymmetry ($p = .024$).

3.6 Publication bias

Table 3. Publication bias diagnostics (exploratory; unreliable under extreme heterogeneity).

Test	Statistic	p	Interpretation
Egger’s regression	intercept = 2.53, $t(39) = 2.34$	.024	Significant funnel asymmetry
Begg’s rank correlation	$\tau = 0.06$	.61	No evidence of bias
Trim-and-fill	0 studies imputed	—	No adjustment

The positive pooled estimate may be inflated by selective reporting; all three diagnostics are unreliable at $I^2 > 90\%$.

3.7 Further sensitivity analyses

Table 4. Sensitivity analyses.

Analysis	k	log-OR	95% CI	τ^2	Approx. RD
Accuracy only (PRIMARY)	37	0.46	[0.22, 0.71]	0.40	≈3 pp
Full sample (+win/resolve-rate)	41	0.45	[0.23, 0.67]	0.34	≈3 pp
Compute-parity only	9	0.31	[0.006, 0.61]	0.08	≈5 pp
Verified / source-located	10	0.35	[0.19, 0.51]	0.01	≈6 pp
2025+ publications only	29	0.41	[0.10, 0.71]	0.47	—
High-RoB removed	16	0.42	[0.13, 0.72]	0.21	—
Unverifiable removed	39	0.43	[0.20, 0.65]	0.34	—
Non-aggregated benchmarks	36	0.48	[0.23, 0.74]	0.42	—
Standard-benchmark denom.	28	0.40	[0.13, 0.67]	0.32	—

Leave-one-out: pooled log-OR range [0.40, 0.48]; most influential omission Wu2025 $\rightarrow$ 0.40 ($\tau^2 = 0.23$).

4 Discussion

4.1 Principal findings

Multi-agent LLM systems appear to outperform single-agent baselines on average (RQ1; log-OR = 0.45), but the absolute margin is ≈ 3 pp aggregate ($h = 0.07$), the prediction interval crosses zero,

and Egger’s test ($p = .024$) indicates likely inflation by small-study effects. The verified subset ($k = 10$, $\log\text{-OR} = 0.35$) provides the most credible anchor, with dramatically less heterogeneity.

4.2 The compute confound

Under compute parity (RQ2) the relative effect shrinks $\sim 31\%$ ($0.45 \rightarrow 0.31$) and the CI barely escapes zero. Zhang et al. (2025) demonstrate that multi-agent debate often fails to beat single-agent chain-of-thought and self-consistency under equal inference compute; Sprague et al. (2024) find that chain-of-thought gains are concentrated in math and symbolic reasoning. A multi-agent debate consuming $5\times$ the tokens of a single agent should be compared against a single agent given the same token budget via chain-of-thought, self-consistency, or repeated sampling. That the relative advantage barely survives compute equalization weakens the practical case for orchestration as a general default.

5 Conclusion

This project does not establish that multi-agent LLM systems generally outperform single-agent baselines. It suggests a small positive average effect ($\log\text{-OR} = 0.45$; aggregate RD ≈ 3 pp, Cohen’s $h = 0.07$), but the evidence is highly uncertain: heterogeneity is extreme, the prediction interval crosses zero, publication bias is detected, and most extraction rows are unverified.

Under compute parity the relative effect declines to $\log\text{-OR} 0.31$. Multi-agent orchestration may help under generous compute budgets, but the evidence base is too contaminated to support strong claims. The field needs compute-controlled comparisons and independent extraction audits.

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Appendix A: Full Extraction Table ($k = 41$)

All 41 study-level comparisons included in the meta-analysis. MA% and SA% are multi-agent and single-agent accuracy; RD is the risk difference in percentage points; log-OR and h are the computed effect sizes.

ID	Citation	Benchmark	Task	Arch.	n	MA%	SA%	RD	log-OR	h	n-src	Compute	Audit	RoB
S001	Du2023	GSM8K	math	debate	1319	85.0	77.0	+8.0	0.524	0.204	exact	no	unaudited	high
S002	Qian2024	SoftwareDev	code	role-play	70	1.4	1.4	0.0	0.000	0.000	imputed	no	unaudited	high
S003	Liang2024	Counter-Intuitive AR	evaluation	cooperation	200	37.0	28.0	+9.0	0.412	0.193	imputed	no	unaudited	high
S004	Huang2024	HumanEval	code	cooperation	164	96.3	90.2	+6.1	1.046	0.250	exact	no	unaudited	high
S005	Tian2025	GPQA-Diamond	other	cooperation	198	86.9	85.9	+1.0	0.086	0.029	exact	no	verified	high
S006	Yang2025	HumanEval	code	cooperation	164	82.3	76.2	+6.1	0.373	0.152	exact	unclear	unaudited	some
S007	Islam2024	HumanEval	code	plan-exec	164	93.9	88.4	+5.5	0.702	0.195	exact	no	ver-corr	high
S008	Hong2024	HumanEval	code	hierarchy	164	86.0	67.1	+18.9	1.102	0.455	exact	no	unaudited	high
S009	Zhang2024b	SWE-bench Lite	code	cooperation	300	34.3	27.3	+7.0	0.329	0.152	exact	unclear	ver-corr	some
S010	Benkovich2026	SWE-bench 500	code	role-play	500	72.0	65.0	+7.0	0.325	0.150	exact	yes	unaudited	some
S011	Xie2025	HumanEval	code	ver-critic	164	70.1	28.7	+41.4	1.765	0.855	exact	no	unaudited	high
S012	Doki2024	HumanEval	code	cooperation	164	98.2	88.4	+9.8	1.950	0.423	exact	no	unaudited	high
S013	Chen2026	ClassEval	code	cooperation	100	82.0	61.0	+21.0	1.069	0.463	imputed	yes	unverifiable	some
S014	Kim2024	MedQA	medical	hierarchy	1273	83.9	79.7	+4.2	0.281	0.109	exact	no	verified	high
S015	Tang2024	MedQA (avg 9)	medical	debate	1273	86.7	80.6	+6.1	0.453	0.170	exact	no	unaudited	high
S016	Li2025	MedXpert-MM	other	ver-critic	581	34.1	28.1	+6.0	0.282	0.130	exact	no	unaudited	high
S017	MediHive2026	MedQA	medical	debate	1273	84.3	77.0	+7.3	0.473	0.187	exact	unclear	unaudited	some
S018	MedARC2025	PubMedQA	medical	debate	1000	77.2	72.9	+4.3	0.230	0.097	exact	no	unaudited	high
S019	Charney2026	Clinical (5 tasks)	medical	hierarchy	100	91.0	73.0	+18.0	1.319	0.483	imputed	yes	unaudited	some
S020	Ophth2026	Ophth. vignettes	medical	debate	200	96.0	86.0	+10.0	1.363	0.364	imputed	no	ver-corr	high
S021	MedLA2025	MedDDx-Expert	medical	debate	570	41.9	31.1	+10.8	0.472	0.225	exact	yes	unaudited	some
S022	Bao2025	MMLU-Pro-Health	gen-know	cooperation	14042	74.4	71.8	+2.6	0.137	0.058	exact	no	unaudited	high
S023	Ferber2025	GI onc. (20 cases)	other	hierarchy	20	85.0	30.0	+55.0	2.582	1.187	exact	no	unaudited	high
S024	Topologies2026	Rare disease (302)	evaluation	hierarchy	302	50.0	48.0	+2.0	0.080	0.040	exact	yes	unaudited	some
S025	Chen2024	SWE-bench Lite	code	hierarchy	300	28.3	18.0	+10.3	0.588	0.239	exact	no	ver-corr	high
S026	Han2025	Weibo21/FakeNews	factuality	debate	4488	82.0	74.0	+8.0	0.470	0.194	estimated	no	verified	high
S027	Fan2025	GSM8K	math	debate	1319	84.8	71.3	+13.5	0.814	0.335	exact	no	unverifiable	high
S028	Wu2025	Knight-Knave-Spy	other	debate	100	69.0	17.0	+52.0	2.386	1.111	imputed	unclear	unaudited	some
S029	Wynn2025	MMLU	gen-know	debate	14042	73.4	82.6	−9.2	−0.543	−0.225	exact	no	unaudited	high
S030	Yao2025	CommonsenseQA	other	debate	1221	87.0	86.0	+1.0	0.084	0.029	exact	no	verified	high
S031	Lin2025	MMLU	gen-know	debate	14042	83.5	76.3	+7.2	0.452	0.181	exact	no	unaudited	high

ID	Citation	Benchmark	Task	Arch.	n	MA %	SA %	RD	log-OR	h	n-src	Compute	Audit	RoB
S032	Cui2025	8 benchmarks	math	debate	5000	64.4	55.7	+8.7	0.365	0.178	imputed	yes	unaudited	some
S033	Wang2025	GPQA	other	debate	198	48.0	42.9	+5.1	0.204	0.102	exact	yes	unaudited	some
S034	Tian2026	MATH	math	debate	5000	80.2	77.8	+2.4	0.145	0.060	imputed	unclear	unaudited	some
S035	Tran2026	FRAMES	other	debate	824	22.0	25.0	−3.0	−0.169	−0.071	estimated	yes	unaudited	some
S036	Li2026	13 datasets	math	debate	5000	67.0	65.3	+1.7	0.076	0.036	imputed	yes	unaudited	some
S037	Hu2025	LLMBar	evaluation	debate	419	82.1	77.1	+5.0	0.310	0.126	exact	unclear	unaudited	some
S038	Liu2026	MATH500/AMC/AIME	math	debate	5000	57.7	52.9	+4.8	0.194	0.096	imputed	unclear	unaudited	some
S039	Zhou2025	HiddenBench (65)	other	cooperation	65	30.8	81.5	−50.7	−2.296	−1.105	exact	no	unaudited	high
S040	Wang2024	AlpacaEval 2.0	other	moa	805	65.1	57.5	+7.6	0.320	0.155	estimated	no	ver-corr	high
S041	Ki2025	NormAd-ETI	other	debate	500	76.0	71.0	+5.0	0.257	0.112	exact	no	ver-corr	high

n-src = raw n-items-source field (exact / imputed / estimated). Compute = compute-parity-flag. Audit = audit-status (ver-corr = verified-corrected; plan-exec = planner-executor; ver-critic = verifier-critic; gen-know = general-knowledge). RoB = informal risk-of-bias rating. S026 (Han2025, factuality) is merged into “other/mixed” for subgroup analysis. **S040 (AlpacaEval 2.0) is a pairwise win-rate and S009/S010/S025 are SWE-bench resolve-rates**; these four are excluded from the primary $k = 37$ analysis.

Appendix B: Search Strings

Table B. Search agent queries.

Agent	Cluster	Representative search strings
1	Multi-agent debate	"multi-agent debate" LLM accuracy benchmark; "LLM deliberation" "single agent" comparison
2	Cooperative / hierarchical	"multi-agent cooperation" OR "hierarchical agents" LLM benchmark; "agent orchestration" evaluation accuracy
3	SWE / code generation	"multi-agent" code generation benchmark SWE-bench OR HumanEval; "agent collaboration" software engineering
4	Medical / scientific	"multi-agent" medical reasoning OR clinical benchmark LLM; "collaborative LLM" scientific QA
5	Critical / null results	"single agent" outperforms "multi-agent" LLM; "scaling single agent" vs debate OR collaboration
6	Frontier models	"frontier model" "multi-agent" diminishing returns OR compute parity; GPT-4 OR Claude "agent comparison"

Searches across arXiv, ACL Anthology, NeurIPS/ICML/ICLR proceedings, and Google Scholar. Date range: January 2023 to May 2026. Not pre-registered.

Appendix C: Exclusion Table

Of 106 records passing screening, 65 were excluded because they lacked extractable multi-agent v.s. single-agent accuracy counts. The excluded records may not be missing at random. **The 41 included studies likely overrepresent positive results.**

Table C. Reasons for exclusion of 65 records.

Exclusion reason	Count	Notes
Only aggregate/composite metrics reported	18	Win rates, Elo scores, or composite indices without item-level breakdown
Single-agent baseline not reported separately	14	Multi-agent results only; comparator inferred or absent
Qualitative or case-study evaluation only	11	No quantitative benchmark data
Different backbone models between MA and SA	8	Fails comparator requirement
Benchmark not identifiable or proprietary	6	Custom evaluations without public benchmark reference
Insufficient detail to compute log-OR	5	Percentages reported but no item counts or benchmark size
Paper not locatable or access restricted	3	arXiv IDs not found or behind paywall without preprint

Appendix D: Cohen’s h as Secondary Effect Size

Cohen’s $h = 2 \arcsin(\sqrt{p_{MA}}) - 2 \arcsin(\sqrt{p_{SA}})$ provides an absolute-scale effect size less sensitive to base-rate inflation than log-OR. Benchmarks: 0.20 = small, 0.50 = medium, 0.80 = large (Cohen, 1988). Values below are the aggregate (summed-proportion) h , emitted by `src/effect_sizes.pooled_absolute_effect`. The unweighted mean runs higher (0.20 for the full sample).

Table D. Aggregate Cohen’s h for key analyses.

Analysis	k	Cohen's h	Interpretation
Full sample	41	0.07	Trivial (unweighted mean 0.20)
Code generation	11	0.23	Small
Compute-parity only	9	0.11	Trivial-to-small
Medical reasoning	7	0.15	Trivial-to-small
General knowledge	3	0.01	No effect